Bihar Engineering University, Patna

B.Tech 5th Semester Exam., 2025

Course: B.Tech ***Time: 03 Hours***
Code: 104501 ***Subject: Computer Networks and Security*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 **Choose the correct option / answer the following (Any seven question only):** **[2 x 7 = 14]**

(a) In computer network which of the following is an example of guided transmission media?
(i) Radio waves (iii) Optical fiber
(ii) Microwave (iv) Infrared

(b) Which techniques is used in Cyclic Redundancy Check?
(i) XOR Encryption (iii) OR operation
(ii) AND operation (iv) Modulo-2 Division

(c) Multiple access protocol which is used carrier sensing and collision detection?
(i) CSMA/CD (iii) Pure ALOHA
(ii) CDMA/CA (iv) Slotted ALOHA

(d) What is the primary function of packet forwarding at the network layer?
(i) Delivering packets to their next hop on the path to the destination (iii) Encrypting data
(ii) Assigning IP addresses (iv) Fragmenting packets

(e) One of the features that is provided by TCP but not by UDP?
(i) Error detection (iii) Broadcasting
(ii) Connection-oriented communication (iv) Fast delivery

(f) Which protocol is used for monitoring and managing network devices?
(i) DNS (iii) SMTP
(ii) SNMP (iv) FTP

(g) A hash function takes an input message and returns .....
(i) a fixed-size string of characters (iii) input size string characters
(ii) a variable-size string of characters (iv) a public-private key pair

(h) What is the main idea behind the RSA public-key encryption algorithm?
(i) Using a single symmetric key (iii) Using large prime numbers and modular arithmetic
(ii) Hiding the IP address of the sender (iv) Using XOR operations on data

(i) The primary function of a Virtual Private Network is?
(i) Creates a secure tunnel over a public network for private communication (iii) Assigns MAC addresses dynamically
(ii) Acts as a physical router (iv) Encrypts DNS records only

(j) In SSL architecture, which protocol is responsible for managing encryption, compression, and message integrity?
(i) SSL Handshake Protocol (iii) SSL Record Protocol
(ii) SSL Alert Protocol (iv) Change Cipher Spec Protocol

Q.2 **Solve both questions :** **[7 x 2 = 14]**

(a) Describe various network topologies used in computer networks.

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(b) Discuss Frequency Division Multiplexing, Time Division Multiplexing, Wavelength Division Multiplexing.

Q.3 **Solve both questions :** **[7 x 2 = 14]**

(a) Explain its features and advantages over TCP and UDP. Discuss the concept of congestion control in SCTP.

(b) Illustrate Logical addressing – IPv4, IPv6.

Q.4 **Solve both questions :** **[7 x 2 = 14]**

(a) Differentiate between wired LAN and wireless LAN. Discuss the concept a Virtual LAN.

(b) Explain the OSI reference model in detail. How it is different from TCP/IP model.

Q.5 **Solve both questions :** **[7 x 2 = 14]**

(a) Explain the working principles of Pure ALOHA, Slotted ALOHA, CSMA/CD, and CSMA/CA with their applications.

(b) Explain the Sliding Window Protocol and Piggybacking technique. How do these methods improve data transmission efficiency in computer networks?

Q.6 **Solve both questions :** **[7 x 2 = 14]**

(a) What is the size of an ARP packet when the protocol is IP and the hardware is Ethernet?

(b) Discuss Three-ways Handshaking in a TCP connection.

Q.7 **Solve both questions :** **[7 x 2 = 14]**

(a) Define Internet Protocol. Is the size of the ARP packet fixed? Explain.

(b) An IP datagram must go through router 128.46.10.5. There are no other restrictions on the routers to be visited. Draw the IP options with their values.

Q.8 **Solve both questions :** **[7 x 2 = 14]**

(a) Discuss Digital Signatures. Explain Secret-key signature and Public-key signature.

(b) The Diffie-Hellman key exchange is being used to establish a secret key between Alice and Bob. Alice send Bob (719, 3, 191). Bob respond with (543). Alice's secret key, x, is 16. Explain the secret key with the help of above example.

Q.9 **Write short notes on any two of the following:** **[7 x 2 = 14]**

(a) Token Bucket algorithm

(b) File Transfer Protocol

(c) Secure Socket Layer and Transport layer Security

(d) Error control protocols

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